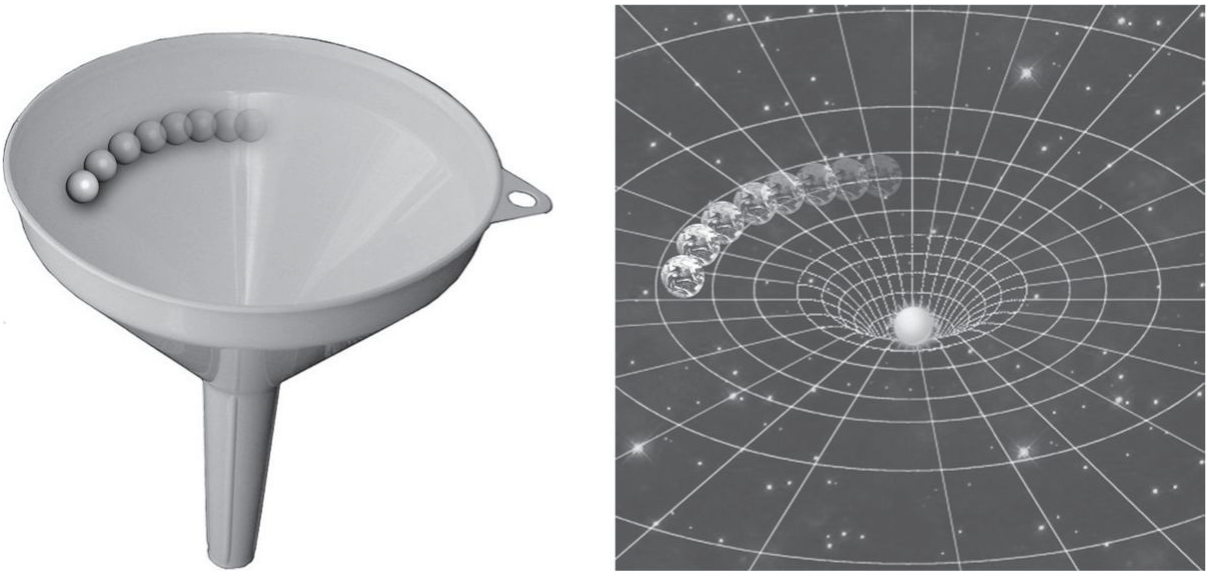


Figure 3.6 The Earth turns around the Sun because spacetime around the Sun is curved, rather like a bead which rolls on the curved wall of a funnel.

A little more precisely, what curves is not space but spacetime – that spacetime which, ten years previously, Einstein himself had shown to be a structured whole rather than a succession of instants.

This is the idea. Einstein's only problem was to find the equations to make it concrete. How to describe this bending of spacetime? And here Einstein is lucky: the problem had already been solved by the mathematicians.

The greatest mathematician of the nineteenth century, Carl Friedrich Gauss, the 'prince of mathematicians', had written maths to describe curved surfaces, such as the surfaces of hills, or such as the one portrayed in figure 3.7.

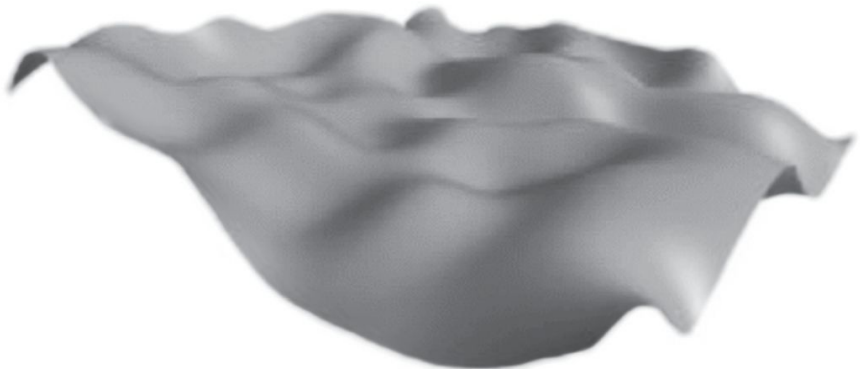


Figure 3.7 A curved (bidimensional) surface.

Then he had asked a talented student of his to generalize this maths to curved spaces in three or more dimensions. The student, Bernhard Riemann, produced a ponderous doctoral thesis of the kind that seems completely useless.